



Guidelines for Management of Margins in Securities Markets and Derivatives Market

Formulated on February 9, 2018

Amended on September 28, 2018

February 15, 2019; March 27, 2019

February 19, 2021; April 16, 2021; May 6, 2021; November 25, 2021

August 16, 2022; December 29, 2022

March 4, 2025; June 4, 2025; October 22, 2025

April 17, 2026; May 19, 2026

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CHAPTER I. GENERAL PROVISIONS

§1. Purpose

The purpose of this Guideline is to stipulate the necessary matters with respect to the methods, procedures and etc. by which the Korea Exchange determines the variables for calculation of margin amount including the margin rates of the Exchange's securities markets and derivatives market pursuant to [§33] and [Annex 5] of the Enforcement Rules of the Securities Market Clearing and Settlement Business Regulation and [§94], [§134] and [Annex 19] of the Enforcement Rules of Derivatives Market Business Regulation. (Amended on February 15, 2019; May 6, 2021; March 4, 2025)

§2. Definitions

(1) The definitions of terms used in this Guideline are as follows: (Amended on February 15, 2019)

1. The term “securities markets” refer to the KOSPI Market, the KOSDAQ Market and the KONEX Market established by the Korea Exchange (hereinafter referred to as “the Exchange”) for the trading of securities pursuant to the [§8-2(4)1] of the Financial Investment Services and Capital Markets Act (hereinafter referred to as “the Act”).
2. The term “derivatives market” refers to the Derivatives Market established by the Exchange for the trading of Exchange-traded derivatives pursuant to the [§8-2(4)2] of the Act.
3. The term “securities type” refers to the stocks, depository receipts of foreign stocks, stocks of real estate investment companies, stocks of ship investment companies, stocks of investment companies, stocks of infra fund companies, listed beneficiary certificates, collective investment securities of exchange-traded funds (including foreign exchange-traded funds; hereinafter the same applies), exchange-traded notes, stocks of investment company for business development collective investment vehicle, investment trust beneficiary certificates of business development collective investment vehicle, equity-linked warrants, subscription warrants, and subscription right certificates. (Amended on April 17, 2026)
4. The term “market index” refers to the indices calculated by the Exchange, such as Korea Composite Stock Price Index (KOSPI) of the KOSPI Market, KOSDAQ Stock

Price Index (KOSDAQ Index) of the KOSDAQ Market, and KONEX Stock Price Index (KONEX Index) of the KONEX Market.

5. The term “sector index” refers to the indices by industrial sector, calculated by the Exchange, of the KOSPI Market and KOSDAQ Market.

6. The term “issue price” refers to the closing price of an issue established in the regular sessions, the special quotation price in the case where there is no transaction matched for the issue on the day, or the base price in the case where there is no special quotation price applicable.

7. The term “margin calculation variables” refers to the variables for calculation of margin amount such as the margin rates of the Exchange’s securities markets (including the transactions at multilateral trading companies; the same hereinafter) and derivatives markets. (Amended on March 4, 2025)

(2) Other terms used in this Guideline shall have the same meanings given to them in the Act and the business regulations of the Exchange except as otherwise provided in this Guideline. (Amended on February 15, 2019)

§3. Basic Principles

(1) The Exchange shall determine margin calculation variables in consideration of the international standards such as the Principles for Financial Market Infrastructures (“PFMIs”) published jointly by the Committee on Payments and Markets Infrastructures of the Bank for International Settlements (CPMI) and the International Organization of Securities Commissions (IOSCO).

(2) The Exchange may determine the margin calculation variables by taking into account various situations such as the characteristics of each product, market situations and objective quantitative information, etc.

CHAPTER II. MARGIN CALCULATION VARIABLES OF SECURITIES MARKETS

§4. Securities Markets Margin Rates

(1) The member margin rates of an issue traded in the securities markets shall be determined for each securities type to which the concerned issue belongs.

(2) The member margin rates by securities type shall be determined by taking into account various factors such as the characteristics of the securities type and market conditions, etc., and adjusted in a timely manner based on the price volatilities of the issues belonging to the concerned securities type or the related indices such as the market index and sector index, etc. (hereinafter referred to as the “related index”).

§5. Measurement of Price Volatility of Securities Markets Issue and Related Index

(1) The price volatility of the issue or the related index noted in [§4(2)], which serves as the basis for the calculation of margin rates shall be measured in accordance with [Annex 1] and [Annex 2]. (Amended on November 25, 2021)

(2) Notwithstanding Paragraph (1), the Exchange may determine the price volatility differently in cases when it deems necessary for market management.
[May 6, 2021]

§6. Calculation of Margin Rates by Securities Type

The member margin rates shall be the value calculated by the method specified in [Annex 3] based on the price volatility pursuant to [§5].

[November 25, 2021]

§7. Regular Change of Securities Market Margin Rates

(1) The Exchange shall change the member margin rates in the securities market on the first Monday (if it falls on a market holiday, it shall be the next trading day; hereinafter referred to as the “regular change date” in this Article) of every month. In such cases, the details of the changes shall be notified to members, etc. at least three (3) trading days prior to the regular change date. (Amended on May 6, 2021; August 16, 2022)

(2) The member margin rates changed in accordance with Paragraph (1) shall be a value obtained based on the price volatility measured on the 7th trading day prior to the regular change date (hereinafter referred to as the “regular calculation base date” in this Article) and decided by the Margin Rates Adjustment Council (hereinafter referred to as the “Council” in this Article). (Amended on August 16, 2022)

(3) The Council meeting for the regular change of the margin rates in the securities market shall be held on the 4th trading day preceding the regular change date. [August 16, 2022]

(4) Notwithstanding Paragraphs (2) and (3), chairperson of the Council may change the regular calculation base date and the Council meeting date if he/she deems it inevitable. [August 16, 2022]

§8. Irregular Change of Securities Market Margin Rates

The Exchange may change the member margin rates by reviewing the adequacy of member margin rates from time to time when it deems necessary for market management due to an abrupt change in the market situation, etc. In such cases, the Exchange shall notify the members, etc. of the details of the change without delay.

§9. Unit of Change of Securities Market Margin Rates

In the case of changing member margin rates pursuant to [§7] and [§8], the minimum increase or decrease unit shall be 0.01 percent point.

CHAPTER III. MARGIN CALCULATION VARIABLES OF DERIVATIVES MARKET

§10. Measurement of Underlying Asset Price Volatility

(1) The price volatility of underlying assets for each product shall be measured by [Annex 1] and [Annex 2] based on the prices of underlying assets. In such cases, the “issue price or closing index” shall be deemed as the “closing price of underlying asset for each product”. (Amended on May 6, 2021; November 25, 2021)

(2) Notwithstanding Paragraph (1), the Exchange may determine the price volatility differently in cases when it deems necessary for market management. (Amended on May 6, 2021) (Moved from previous Paragraph (3), and previous Paragraph (2) deleted on November 25, 2021)

(4) [Deleted on May 6, 2021]

(5) [Deleted on May 6, 2021]

(6) [Deleted on May 6, 2021]

§11. Calculation of Derivatives Market Margin Rates

The member margin rates shall be the value calculated by the method specified in [Annex 3] based on the price volatility pursuant to [§10].

[November 25, 2021]

§12. Regular Change of Derivatives Market Margin Rates

(1) The Exchange shall change the member margin rates in the derivatives market on the first Monday (if it falls on a market holiday, it shall be the next business day; hereinafter referred to as the “regular change date” in this Article) of every month. In such cases, the details of the changes shall be notified to members, etc. at least three (3) business days prior to the regular change date. (Amended on May 6, 2021; August 16, 2022; June 9, 2025)

(2) The member margin rates changed in accordance with Paragraph (1) shall be a value obtained based on the price volatility measured at the time of market closing on the 7th business day prior to the regular change date (hereinafter referred to as the “regular calculation base date” in this Article) and decided by the Margin Rates Adjustment Council (hereinafter referred to as the “Council” in this Article). (Amended on August 16, 2022; June 9, 2025)

(3) The Council meeting for the regular change of the margin rates in the derivatives market shall be held on the 4th business day preceding the regular change date. [August 16, 2022] (Amended on June 9, 2025)

(4) Notwithstanding Paragraphs (2) and (3), chairperson of the Council may change the regular calculation base date and the Council meeting date if he/she deems it inevitable. [August 16, 2022]

§13. Irregular Change of Derivatives Market Margin Rates

The Exchange may change the member margin rates by reviewing the adequacy of member margin rates from time to time when it deems necessary for market management due to an abrupt change in market situation, etc. In such cases, the Exchange shall notify the members, etc. of the details of the change without delay.

§14. Temporary Adjustment of Derivatives Market Margin Rates during Holidays

(1) In cases where the market holidays continue for three (3) consecutive days or longer and the Exchange deems it necessary for market management, the member margin rates may be temporarily adjusted. (Amended on February 15, 2019; June 9, 2025)

(2) When making a temporary adjustment of the member margin rates during holidays, the Exchange shall notify the members of the adjustment schedule and the extent of adjustment by 10 business days and five (5) business days prior to the adjustment date, respectively. (Amended on June 9, 2025)

(3) The temporary adjustment of member margin rates during holidays shall not apply to the calculation of the customer margins and the credit risk loss equivalent under [§94-2(1)] of the “Enforcement Rules of Derivatives Market Business Regulation”. (Amended on February 15, 2019)

§15. Unit of Change

In the case of changing member margin rates, the minimum change unit shall be 0.10 percent point for the index and equity derivatives, 0.01 percent point for the interest rate derivatives and the currency derivatives and 0.10 percent point for the commodity derivatives.

§16. Customer Maintenance Margin Rates and Customer Margin Rates

(1) The rates of the customer maintenance margins shall be the same as the rates of the member margins.

(2) The customer margin rates shall be set at 1.5 times the customer maintenance margin rates. However, when it is deemed necessary for market management, the Exchange may set it differently. (Amended on May 6, 2021)

§17. Composition of Product Group

The Exchange shall establish product groups to which the margin reduction is applied in accordance with [Annex 4] by comprehensively considering the quantitative analysis of correlations among underlying assets’ rates of return (referring to the value obtained by dividing the difference between the closing prices of the underlying asset on the concerned day and the day before by the closing price of the day before; hereinafter the same shall apply in this Chapter) and the qualitative factors such as the similarity of underlying assets’ characteristics and the industry sectors, etc.

[Wholly amended on December 29, 2022]

§18. Determination of Price Correlation Rates

The Exchange shall set the price correlation rate of a product group in accordance with [Annex 4] by considering the level of movement to the same direction of the rates of return among underlying assets in the same product group.

[Wholly amended on December 29, 2022]

§19. Relative Scale Rates

The Exchange shall establish the relative scale rates in accordance with [Annex 4] by considering the multipliers and the prices of underlying assets of each product in the same product group.

[Wholly amended on December 29, 2022]

§20. Change of Variables for Calculation of Margin Reduction Amount

(1) The Exchange shall review the adequacy of the composition of product groups, the price correlation rates, and the relative scale rates (hereinafter referred to as the “reduction amount calculation variables”) twice a year and change the reduction amount calculation variables when it is deemed necessary. However, the Exchange may not conduct the review of the adequacy of the reduction amount calculation variables in cases where the change in the market situation is insignificant.

(2) The Exchange shall notify the details of the change to the members, etc. at least by three (3) business days prior to the application date of the change in the reduction amount calculation variables. (Amended on June 9, 2025)

§21. Futures Spreads Margin Rates

The member margin rates of futures spreads shall be set considering the differences in price movements among delivery month futures contracts, price levels of the underlying assets, and the characteristics of products, etc.

§22. Minimum Margin per Contract

The minimum member margin per contract shall be calculated considering the cost of spreads, etc. for the disposition of open interests.

CHAPTER IV. MANAGEMENT OF BUSINESS RELATING TO MARGIN CALCULATION VARIABLES

§23. Operation of Council

(1) The Margin Rates Adjustment Council (hereinafter referred to as “the Council”) shall be in operation for the deliberation of matters on review and change of margin rates of the securities and derivatives markets.

(2) The Council shall be composed of a chairperson and council members as classified in each of the following subparagraphs and have a secretary in charge of handling the matters on council business including preparation of meeting materials. (Amended on September 28, 2018; February 15, 2019; March 27, 2019; April 16, 2021; May 6, 2021)

1. Chairperson who is the officer in charge of CCP risks pursuant to [§7-2(2)] of the Risk Management Committee Operation Regulation;
2. Council members who are the heads of the department in charge of clearing and settlement, CCP risk management department, and CCP risk assessment department; and
3. Secretary who is the head of the team in charge of margin-related tasks.

(3) The Council may take opinions from the heads of the following departments or outside experts when necessary: (Amended on May 6, 2021)

1. Securities market: Heads of departments related to system and operational affairs of the securities markets; and
2. Derivatives market: Heads of departments related to system and operational affairs of the derivatives market.

(4) Notwithstanding the provisions of [§6] and [§11], the Council may set the margin rates by comprehensively considering such factors as the domestic and global market situations, outlook on the price volatilities and influence on the settlement stability, etc. (Amended on March 4, 2025)

(5) Notwithstanding Paragraph (1), the Council meeting may be replaced by a written meeting in consideration of the scope of adjustment of the member margin rates, the number of products subject to member margin rates change, etc., and in the case of an inevitable situation such as an urgent irregular change of the member margin rates, the meeting of the Council may be omitted. (Amended on March 27, 2019)

§24. Management of Margin Calculation Variables

The matters necessary for the determination of the margin calculation variables including those not stipulated in this Guideline shall be determined by the head of the department in charge of managing the margin calculation variables.

§25. Assessment of Methods for Determining Securities Market Margin Rates

(Amended on May 6, 2021)

(1) The Exchange shall periodically check the appropriateness of the methods for determining the member margin rates in the securities markets through ex-post verification of whether the amount of the securities markets member margin per clearing member is sufficient to compensate the actual losses.

(2) In order to assess the adequacy of Paragraph (1), the Exchange may, in respect of each trading day, calculate the number of days in which the loss amount (it shall be the loss equivalents incurred as of the market closing times of the concerned day and two (2) trading days thereafter) of each clearing member exceeds the net exposure margin amount of the securities markets during the immediately preceding 250 trading days and change the methods for determining the member margin rates in consideration of the foregoing number of days calculated in cases where it deems necessary for market management. (Amended on May 6, 2021)

§26. Assessment of Methods for Calculating Derivatives Market Margin Rates

(1) The Exchange shall periodically check the appropriateness of the methods for determining the member margin rates of the derivatives market through an ex-post verification of whether the amount of the derivatives market member margin per clearing member is sufficient to compensate the actual losses. (Amended on May 6, 2021)

(2) Pursuant to Paragraph (1), the Exchange may change the methods for determining the member margin rates in cases where it deems necessary for market management considering the number of days in which the loss amount (it shall be the loss equivalents incurred as of the market closing times of the concerned day and two (2) trading days thereafter) of each clearing member exceeds the member margin amount of the derivatives market during the immediately preceding 250 trading days in respect of each trading day. (Amended on May 6, 2021)

(3) [Deleted on May 6, 2021]

§27. Public Notification

(1) The Exchange shall publicly notify the methods and procedures for managing margin calculation variables and the details of changes via its website, etc.

(2) When changing the margin calculation variables, the Exchange shall notify the details of the change to the members without delay, and publicly notify them via its website and the market operation terminals, etc.

ADDENDA

(Regulation No. 589, February 9, 2018)

§1. Effective Date

This Guideline shall become effective on February 12, 2018. However, the provisions of [§12(1)] shall become effective on January 2, 2019 and the provisions of [§16] shall become effective on the date that the Exchange sets separately considering the time period required for the development of computer programs, etc.

§2. Interim Measures concerning Regular Change of Derivatives Market Margin Rates

In applying the provisions of [§12(1)], the review of the adequacy of margin rates shall be made based on the 15th day of each even-numbered month until December 28, 2018 from the effective date of this Guideline.

ADDENDUM

(Regulation No. 613, September 28, 2018)

This Guideline shall become effective on October 1, 2018.

ADDENDUM

(Regulation No. 633, February 15, 2019)

This Guideline shall become effective on February 18, 2019.

ADDENDUM

(Regulation No. 639, March 27, 2019)

This Guideline shall become effective on March 29, 2019.

ADDENDUM

(Regulation No. 717, February 19, 2021)

This Guideline shall become effective on February 19, 2021.

ADDENDUM
(Regulation No. 726, April 16, 2021)

This Guideline shall become effective on April 16, 2021.

ADDENDUM
(Regulation No. 734, May 6, 2021)

This Guideline shall become effective on May 7, 2021.

ADDENDA
(Regulation No. 756, November 25, 2021)

§1. Effective Date

This Guideline shall become effective on November 29, 2021

§2. Exceptional Cases concerning Application of Procyclicality Mitigation Rate in Calculation of Base Member Margin Rate for the Securities Markets.

Notwithstanding the amended provision of [Annex 3] of this Guideline, the procyclicality mitigation rate applied when calculating the member margin rate for the securities markets shall be adjusted according to each of the following subparagraphs:

1. From the effective date of this Guideline to June 30, 2022, the rate shall be 5 percent;
2. From July 1, 2022 to December 30, 2022, the rate shall be 15 percent; and
3. From January 2, 2023 to December 29, 2023, the rate shall be 20 percent.

§3. Exceptional Cases concerning Buffer in Calculation of Minimum Margin Rate

Notwithstanding the amended provision of [Annex 3] of this Guideline, the buffer applied when calculating the minimum margin rate shall be adjusted according to each of the following subparagraphs:

1. From the effective date of this Guideline to December 30, 2022, the buffer shall be 15 percent; and
2. From January 2, 2023 to December 29, 2023, the buffer shall be 20 percent.

ADDENDUM
(Regulation No. 787, August 16, 2022)

This Guideline shall become effective on August 19, 2022.

ADDENDUM
(Regulation No. 814, December 29, 2022)

This Guideline shall become effective on December 30, 2022.

ADDENDUM
(Regulation No. 920, March 4, 2025)

This Guideline shall become effective on March 4, 2025.

ADDENDUM
(Regulation No. 931, May 30, 2025)

This Guideline shall become effective on June 9, 2025.

ADDENDUM
(Regulation No. 960, October 22, 2025)

This Guideline shall become effective on October 24, 2025.

ADDENDA
(Regulation No. 997, April 17, 2026)

§1. Effective Date

These Guidelines shall become effective on April 27, 2026

§2. Exceptional Cases concerning Member Margin Rate for Business Development Collective Investment Vehicle Investment Company Stocks and Investment Trust Beneficiary Certificates

Notwithstanding the amended provision of [Annex 3] of this Guideline, the member margin rate for the business development collective investment vehicle investment company stocks

and investment trust beneficiary certificates shall be 50% for up to 6 months from the listing date of the first listed issue by security type.

ADDENDUM
(Regulation No. 1004, May 19, 2026)

This Guideline shall become effective on May 26, 2026.

[Annex 1] Price Volatility Measured by Exponentially Weighted Moving Average

(Amended on May 6, 2021; November 25, 2021; May 19, 2026)

Price Volatility Measured by Exponentially Weighted Moving Average

(Pertaining to [§5] and [§10])

$$\sigma_t = \sqrt{\lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2}$$

λ : Impact extinction coefficient. In such cases, the impact extinction coefficient shall be 0.97, but it may be set differently if the Exchange deems it necessary for market management.

σ_{t-1} : Price volatility of the immediately preceding trading day. However, the initial value of the price volatility, which is σ_0 , shall be the standard deviation of the rates of return during the period of 250 trading days from the immediately preceding 501st trading day.

r_{t-1} : Rate of return for the immediately preceding trading day. In such cases, the rate of return shall be calculated by dividing the value obtained by deducting the closing price of the concerned issue or index on the two (2) trading days before the concerned trading day from the closing price of the concerned issue or index on the concerned trading day by the closing price of the concerned issue or index on the two (2) trading days before the concerned trading day.

[Annex 2] Average Price Volatility

[November 25, 2021]

Average Price Volatility (Pertaining to [§5] and [§10])

$$\text{Average price volatility } (\sigma_{avg,t}) = \sqrt{\sum_{i=t-N+2}^{t-1} \{(r_i - \bar{r})^2 / (N - 3)\}}$$

$$\bar{r} = \sum_{i=t-N+2}^{t-1} \{r_i / (N - 2)\}$$

N : Number of trading days in the past ten (10) years from the immediately preceding trading day

r_{t-n} : The rate of return before n trading days. In such cases, the rate of return shall be calculated by dividing the value obtained by deducting the closing price of the concerned issue or index on the two (2) trading days before the concerned trading day from the closing price of the concerned issue or index on the concerned trading day by the closing price of the concerned issue or index on the two (2) trading days before the concerned trading day.

$\bar{r}$: The average rate of return for the past ten (10) year from the immediately preceding trading day.

[\[Annex 3\] Calculation of Margin Rates of Securities Markets and Derivatives Market](#)

[November 25, 2021] (Amended on March 4, 2025; October 22, 2025; April 17, 2026, May 19, 2026)

Calculation of Margin Rates of Securities Markets and Derivatives Market
(Pertaining to [§6] and [§11])

1. Calculation of Securities Market Margin Rate

Security type	Issue	Member margin rate
Stocks listed on the stock markets	Issue with a sector index to which the issue belongs	The larger value of the base member margin rate and the minimum margin rate of the sector index to which the concerned issue belongs
	Issue with no sector index to which the issue belongs	1.2 times the larger value of the base member margin rate and the minimum margin rate of the market index to which the concerned issue belongs
Depository receipts of foreign stocks	-	
Exchange-traded funds and exchange-traded notes	Issue that has been listed for at least six (6) months	The larger value of the base member margin rate and the minimum margin rate of the concerned issue
	Issue that has not been listed for six (6) months	The average value of the base member margin rates of the top 30% issues in terms of trading value (including trading value at multilateral trading companies)
Stocks of real estate investment companies Stocks of ship investment companies Stocks of investment companies Stocks of infra fund companies Listed beneficiary certificates Business development collective investment vehicle investment	-	

stocks and investment trust beneficiary certificates		
Equity-linked warrants	Issue whose underlying asset is KOSPI200	50%
	Issue whose underlying asset is not KOSPI200	60%
Subscription warrants	-	60%
Subscription warrant certificates	-	30%
Non-monetary trust beneficiary securities	-	50%
Investment contract securities		

Notes:

1. ‘The base member margin rate’ is the sum of (i) the value obtained by applying t-distribution (a degree of freedom of 250 and a confidence level of 99.7%) to the average of the price volatilities calculated pursuant to [Annex 1] for each issue or related index subject to calculation; and (ii) this value multiplied by the procyclicality mitigation rate.
2. ‘The minimum margin rate’ is the sum of the value obtained by applying the standard normal distribution (a confidence level of 99.7%) to the price volatility calculated pursuant to [Annex 2] for each issue or related index subject to the calculation and this value multiplied by the buffer.
3. ‘The procyclicality mitigation rate’ and ‘the buffer’ shall be 25 percent, respectively. However, the Exchange may set it differently when it is deemed necessary for market management.

2. Calculation of Derivatives Market Margin Rate

The member margin rate of the derivatives market shall be the larger value of the base member margin rate and the minimum margin rate calculated for each underlying asset or the relevant product by applying Subparagraph 1 *mutatis mutandis*.

3. Exceptions to Calculation of Margin Rates in Securities and Derivatives Markets

Notwithstanding subparagraph 1 and 2, if data for calculating the price volatility of stocks, indices, and underlying assets is insufficient or if it is deemed necessary for market management, the member margin rate may be set differently to reflect the characteristics of the stocks, indices, and underlying assets.

[Annex 4] Calculation of Variables for Margin Reduction Amount in Derivatives Market

[December 29, 2022]

Calculation of Variables for Margin Reduction Amount in Derivatives Market

(Pertaining to [§17], [§18] and [§19])

1. Composition of Products, etc.

A. Product group subject to calculation of margin reduction amount

The product groups for calculating the margin reduction amount are classified as follows based on the underlying assets of the derivatives:

- (1) Interest rate product group;
- (2) Currency product group;
- (3) Stock index product group;
- (4) Equity product group (categorized by industry); and
- (5) Commodity product group.

B. Selection of Underlying Assets by Product Group

When the Exchange conducts the regular review of variables for calculating the margin reduction amount pursuant to [§20(1)], it shall select the underlying assets belonging to the relevant product group in the following order:

- (1) To compose the underlying assets belonging to each product group considering the market classification under [§3] of the Derivatives Market Business Regulation and the industry classification method under the Global Industry Classification Standard (GICS). Provided that, if there is only one (1) underlying asset in a product group, the relevant product group is excluded from the calculation of margin reduction amount;
- (2) To set the sample period corresponding to the 250, 500, and 750 trading days immediately before the review date of variables for calculating margin reduction amount; and

(3) To calculate, for each underlying asset (X), the same directional probability with other underlying assets (Y) in the same product group for each sample period according to the following formula.

$$\text{Same directional probability} = \frac{1 + \tau(X, Y)}{2}$$

$$\tau(X, Y) = \pi_c - \pi_d$$

$$\pi_c = \Pr[(X_i - X_j)(Y_i - Y_j) > 0] \quad (i < j)$$

$$= \frac{\text{The number of frequencies which the price changes } X \text{ and } Y \text{ are the same}}{\frac{n(n-1)}{2}}$$

$$\pi_d = \Pr[(X_i - X_j)(Y_i - Y_j) < 0] \quad (i < j)$$

$$= \frac{\text{The number of frequencies which the directions of price changes } X \text{ and } Y \text{ are the same}}{\frac{n(n-1)}{2}}$$

X_i, Y_i : i -th closing price of underlying assets X and Y during the sample period

(4) To exclude the underlying assets whose average value of the same directional probability calculated according to (3) above during the sample period is less than 0.6 from the product group of the sample period. Provided that an underlying asset of a derivative product, whose average daily trading value over the preceding 250 trading days is in the top 50% or higher within the same product group, and an underlying asset, which is determined that it is a major underlying asset within the industry group including the top two (2) stocks in market capitalization within the industry, may not be excluded; and

(5) To calculate the margin reduction amount noted in (2) and (3) above for the product group of the sample period with the largest number of underlying assets for each product group.

2. Calculation of Price Correlation Rate of Product Group

A. Price correlation between underlying assets

The price correlation rate between the underlying assets (X, Y) within the same product group selected pursuant to paragraph 1 shall be calculated according to the following formula.

$$\text{Price correlation rate} = 1 - \frac{|r_{x,t} - r_{y,t}| \odot 99th\text{quartile}}{\max|r_{x,t}| + \max|r_{y,t}|}$$

$$r_{x,t} = \frac{X_{t+2} - X_t}{X_t}, r_{y,t} = \frac{Y_{t+2} - Y_t}{Y_t}$$

$\max|r_{x,t}|$: Maximum value of $|r_{x,t}|$ during the sample period*

$\max|r_{y,t}|$: Maximum value of $|r_{y,t}|$ during the sample period

* The sample period refers to the sample period in paragraph 1B(5) applied to the product group to which the underlying asset belongs.

B. Price correlation rate of each product group

The price correlation rate of each product group is calculated by truncating the minimum value of the price correlation rate within the product group calculated according to item A by 5%.

3. Calculation of Relative Scale Rate

A. Average contract value of futures product by underlying asset

The average contract value of futures products for each underlying asset is calculated by multiplying the average price of the underlying asset by the trading multiplier of the futures product.

$$\begin{aligned} &\text{Average contract value of futures products by underlying asset} \\ &= \text{Trading multiplier of futures product of underlying asset}^* \times \text{average price}^{**} \text{ of} \\ &\quad \text{underlying asset} \end{aligned}$$

* In the case where there are multiple futures products for an underlying asset, the larger multiplier is applied.

** The average price shall be the average value during the sample period in paragraph 1B(5) applied to the product group to which the relevant underlying asset belongs.

B. Relative scale rate by underlying asset

The relative scale rate by underlying asset is calculated by dividing the largest value of the average contract value of futures products for each underlying asset within the product group to which the underlying asset belongs by the average contract value of futures products of the underlying asset, and in such case, it is rounded off to the 2nd decimal place.

Relative scale rate by underlying asset

=Maximum value of the average contract value of futures products for each underlying asset within the product group/average contract value of futures products of the underlying asset